* Components of a Data Communication System:

Data Communication is defined as exchange of data between two devices via some form of transmission media such as a cable, wore or it can be air or vacuum also.

. The term "Data Communication" comprises two words: Data and Communication. Data can be any text, image, audio, video, and multimedia files. Communication is an act of Sending or receiving data.

. Five most important aspects of data communication are sender, receiver, communication medium, the message to be communicated and certain rules called protocols to be followed during communication.

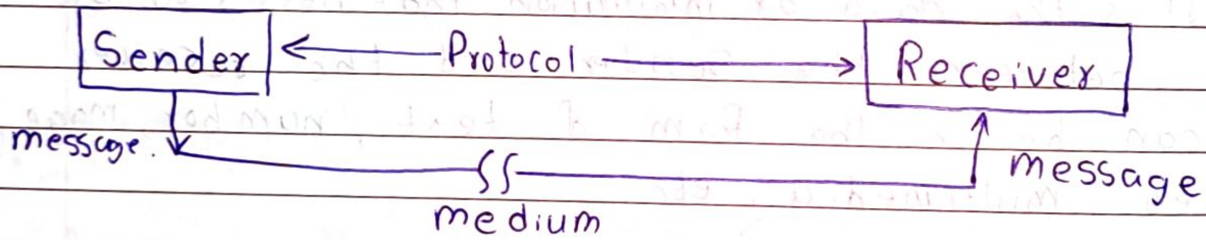
1. Message: It is the data or information that needs to be exchanged between the sender and the receiver. Messages can be in the form of text, number, image, audio, video, multimedia, etc.

2. Communication Media: It is the path through which the message travels between source and destination. It is also called medium or link which is either wired or wireless.
ex. television cable, telephone cable, ethernet cable, satellite link, microwaves, etc.

3. Protocols: It is a set of rules that need to followed by the communicating parties in order to have successful and reliable data communication
ex. HTTP etc., SMTP

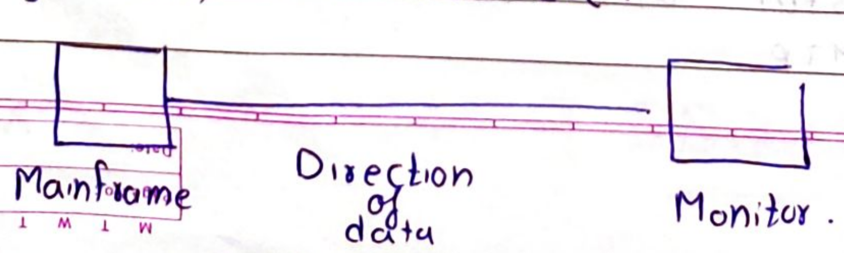
4. Sender: A Sender is a computer or any Such device which is capable of Sending data over a network. It can be a computer, mobile phone, smartwatch, walkie-talkie, video recording device etc. It plays the part of a Source in data communication

5. Receiver: A receiver is a computer or any Such device which is capable of receiving data from the network.
It can be any computer, printer, laptop, mobile, television etc.
In computer communication, the sender and receiver are known as nodes in a network
It is the destination where finally message sent by Source arrives.



Data Flow:
Communication between two devices can be Simplex, half-duplex, or full-duplex

1. Simplex: In Simplex mode, Sender can send data but the sender is unable to receive data.
It is a type of communication in which communication happens in one direction only
ex. keyboard, monitor etc.

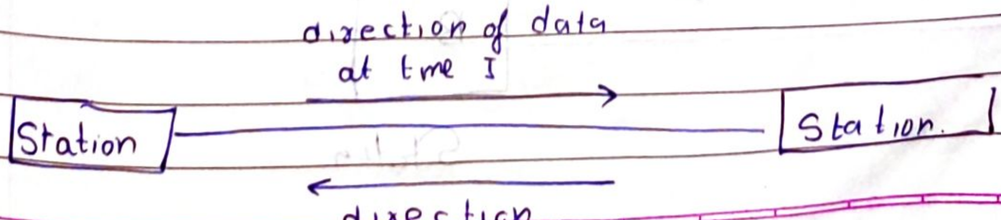


The simplex mode can use the entire capacity of the channel to send data in one direction

* Advantages
- Simplicity: Simplex mode is simple to implement because data travels in only one direction
- Cost-Effective: Since communication is single directional, the hardware required can be less costly.
- No collision: As data travel is single direction, there is no risk of data collision.

* Disadvantages:
- Lack of bidirectional communication: Incapacity to send data back in the opposite way.
- Inefficiency for complex tasks:
- Limited flexibility: It lacks flexibility because they cannot be easily adjusted to situations where bidirectional communication might become necessary.
- Not ideal for modern Networks.

2). Half-Duplex Mode:
In half-duplex mode, sender can send the data and also receive the data one sequentially. It is a bidirectionally communication but limited to only one at a time.
ex. Walkie-talkie in which information is sent one at a time but in bi-directions.



direction of data at time 2.

When one device is sending the other can only receive and vice versa

The entre capacity of the channel can be utilized for each direction

★ Advantages:

- Efficient use of channels : half-duplex allows bidirec communication over a single channel, making it effective for Scenarios where parallel transmission isn't requred.
- Cost effective as compare to full duplex systems
- Simplified Collision handling : Crashes are reduced as one device communicates at a time

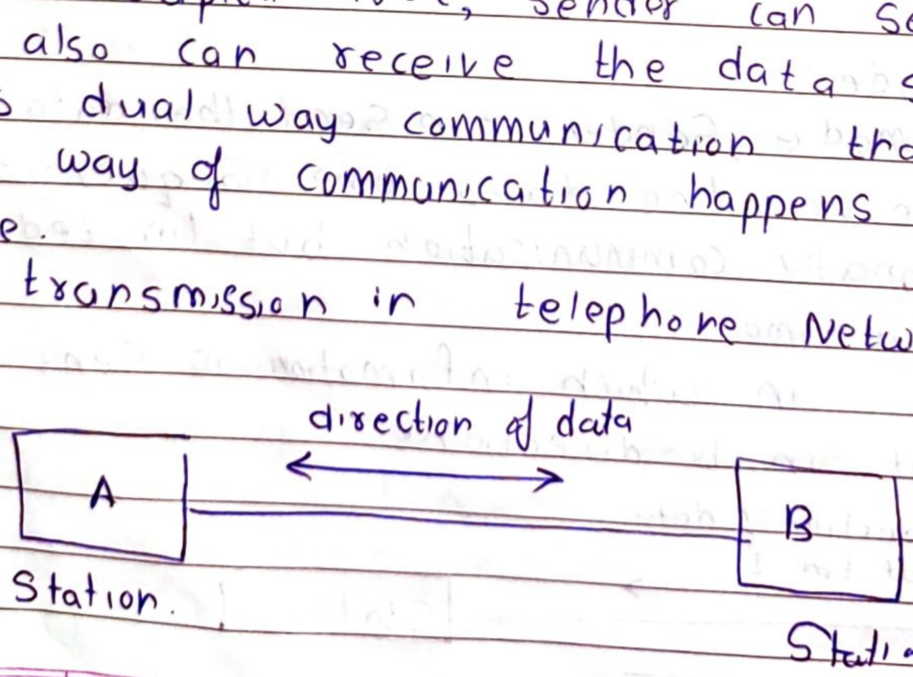
★ Disadvantages:

- Slower data transmission : Communications are delayed as data flows in one direction at a time.
- Increased latency: Changing modes from sending to receiving introduces delays.
- Not ideal for high traffic networks.

★ Full Duplex Mode:

In full-duplex mode, Sender can send the data and also can receive the data simultaneously. It is dual way communication that is both way of communication happens at a same time.

ex - transmission in telephone Networ.



In full-duplex mode, signals going in one direction share the capacity of link.
Sharing can occur in two ways.
- Two separate physical transmission path
- Capacity of channel is divided between signals travelling in both directions.

★ Network:
A network is a set of devices (often referred as nodes) connected by communication links.

★ Physical Structures.
There are two possible types of connections:

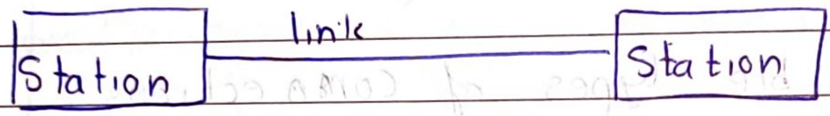
1. Point-to-Point → To be cont. forward.

★ Advantages of full-Duplex Mode:
- High Speed Communication: Data transfer is quicker because there is no delay for channel to clear.
- Reduced latency: Since communication is parallel, time lag is minimized.
- Better Utilization of bandwidth: Capacity is used more efficiently.

★ Disadvantages.
- Complexity and cost: It requires complex hardwares.
- Requires quality Infrastructure: It requires better cabling and more refined networking equipment to avoid interference.

1. Point - to - Point Connection:
- A point to point connection provides a dedicated link between two devices.
- The entire capacity of the link is reserved for transmission between those two devices.
- Most P2P connections use actual wire or caples to connect two ends, but other options, such as microwave or satellite links, are also possible.

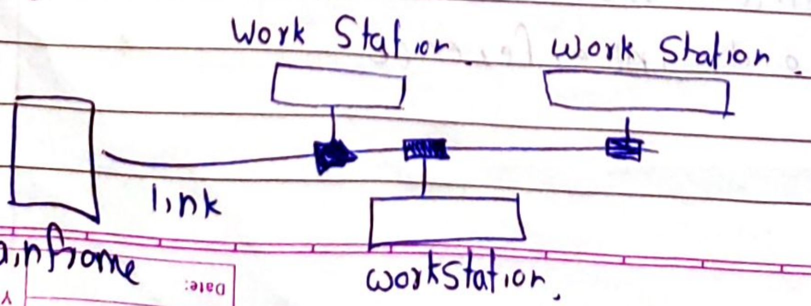
ex. infrared remote control and television.



2. Multipoint Connection:
- It is also called multdrop configuration. In this connection, two or more devices share a single link.
- If more than two devices share the link then the Channel is considered a 'shared channel' with shared capacity, there can be two possibilities in a multipoint line configuration.

i Spacial Sharing: If several devices can share the link simultaneously.

ii Temporal (Time) Sharing: If user must take turns using the link.

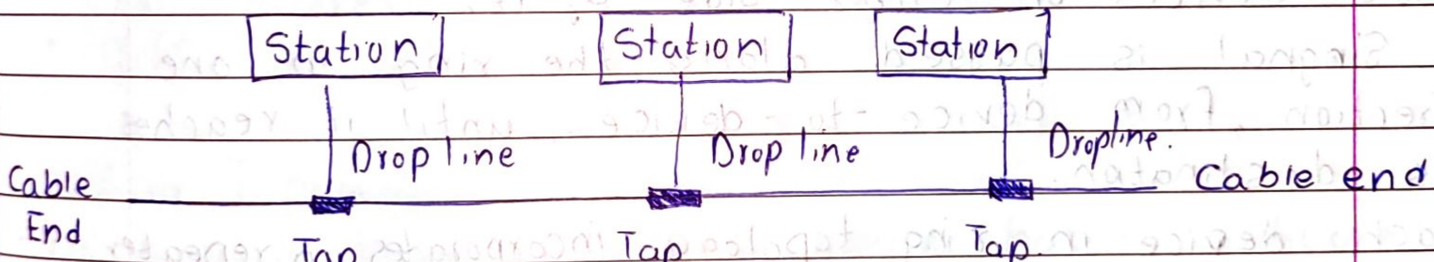


* Physical Topology:

- The term physical topology refers to the way in which a network is laid out physically.
- Two or more devices connect to a link, two or more links form a topology.
- The topology of a network is the geometric representation of the relationship of all the links and linking devices (usually called nodes) to one another.
- There are four basic topologies possible.

1. Bus topology:

- A bus topology is multipoint
- One long cable acts as a backbone to link all the devices in a network.



- It is non robust topology because if the backbone fails the topology crashes.

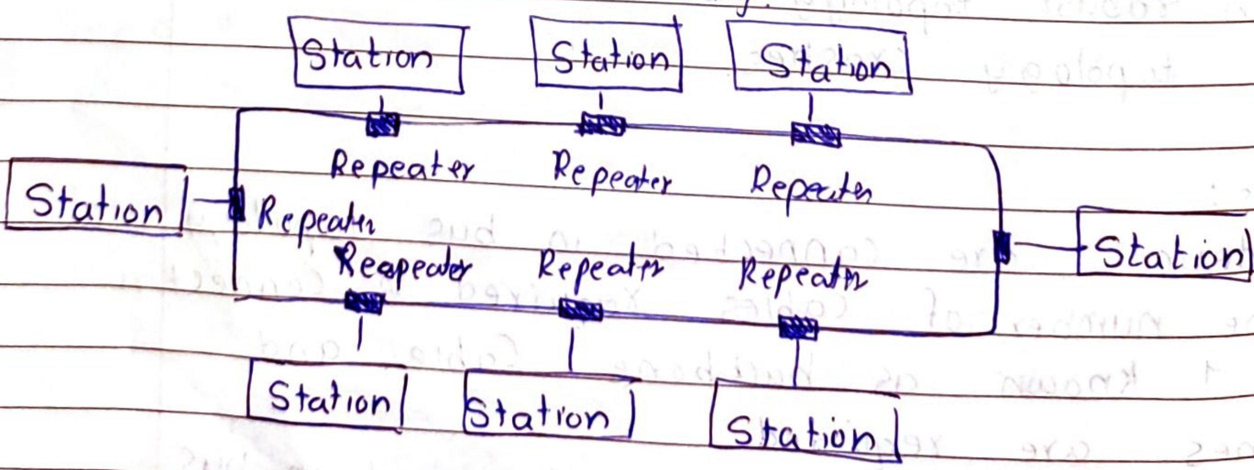
* Advantages:

- If N devices are connected in bus topology then the number of cables required to connect them is 1 known as backbone cable, and N drop lines are required
- Coaxial or twister pair cable are mainly used in bus topology with support upto 10 Mbps.

- The cost of the cable is less compared to other topologies
- Easy to manage and setup.

* Disadvantage
- If common cable fails, then the whole System will crash.
- If network traffic is more, it increases collisions in the network.
- Adding new nodes slow down the network.
- Security is very low.

2) Ring Topology:
- In a ring topology, each device has a dedicated point-to-point connection with only the two devices on either side of it.
- A Singnal is passed along the ring in one direction, from device-to-device, until it reaches its destination.
- Each device in ring topology incorporates a repeater.
- When a device receives a Signal intended for another device, its repeater regenerates the bits and passes them along.

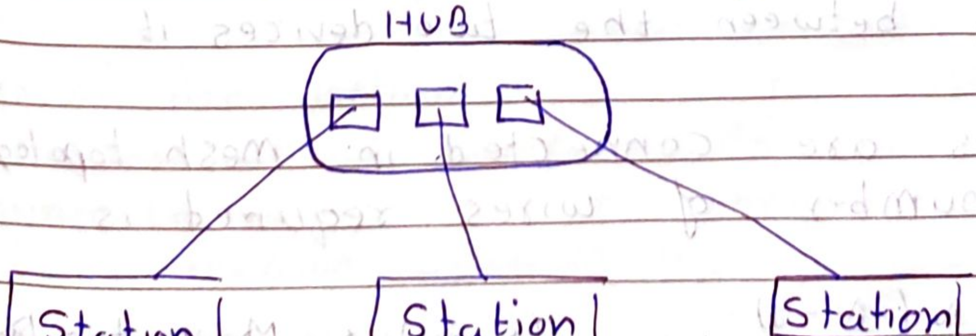


- A ring is relatively easy to install and reconfigure.
- Token Ring Passing protocol is used by the workstation to transmit the data.
- It can be made bidirectional by having 2 connections between each network node.

* Advantages:
- high Speed data transmission.
- Possibility of collision is minimum.
- Cheap to install and extend
- less costly than star topology.

* Disadvantages of Ring Topology:
- Failure of Single node in network can cause entire network to fail
- less Secure.
- Difficult to troubleshoot.

3) Star Topology:
- In a star topology, each device has a dedicated point-to-point link only to a central controller, usually a hub.
- The devices are not directly connected to one another.
- Controller acts as an exchange.



- Coaxial cables or RJ-45 cables are used to connect computers.
- It is used in LANs

* Advantages of Star Topology:
- If N devices are connected, then number of cables required is N.
- New devices can be easily added if there are empty ports in hub.
- Easy to fault identification and fault isolation
- Cost-effective
- Failure of one node does not affect other nodes.

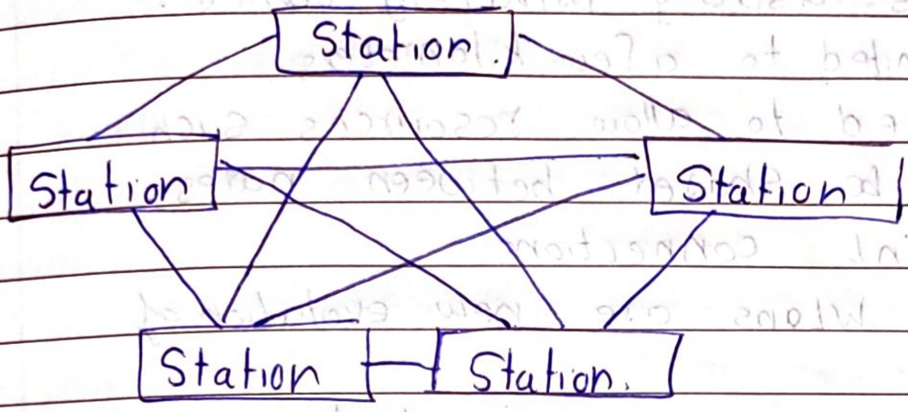
* Disadvantages:
- The cost of initial setup is high.
- It is subjected to single point failure. If central hub fails entire network collapses.
- If hub has limited ports, adding new nodes require upgrading hub.

4) Mesh Topology:
- In mesh topology, every device has a dedicated point-to-point link to every other device.
- The term dedicated means that the link carries traffic only between the two devices it connects.
- If N nodes are connected in mesh topology then the number of wires required is given by

$$\frac{N(N-1)}{2}$$ ~~(Simplex Mesh topology)~~

If each physical link allows communication in both direction

• Each device must have n-1 input/Output (I/O) ports to connect to other n-1 stations.



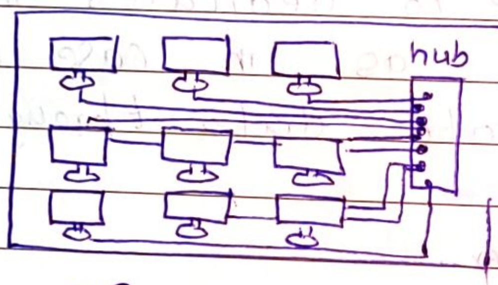
* Advantages.
• Fast communications due to dedicated links
• Mesh topology is robust as in case of error in any link we can rote data through other path.
• Fault diagnose is easier.
• Provides Security and privacy.
• If one link fails it does not affect others.
• Physical boundaries prevent other users from gaining access to message.

* Disadvantage.
• High cost of setup and management
• Complex Configuration due to higher wireing
• Higher number of ports (n(n-1)) are required.

* Types of Area Network.
The network allows computers to connect and communicate with different computers (devices) via any medium. LAN, MAN, WAN are the three major types of networks designed to operate over the area they cover.

1) Local Area Network (LAN):

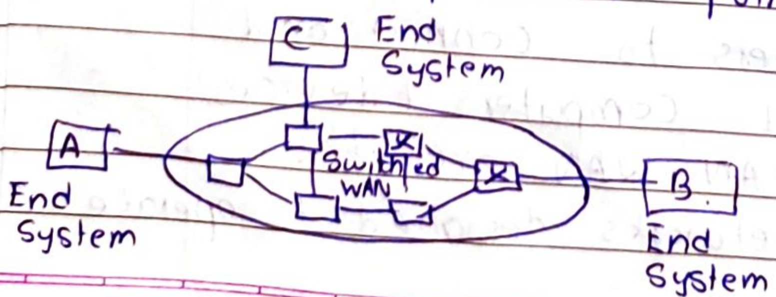
- LAN or Local Area Network connects network devices within a single office, building or campus and is usually privately owned.
- LAN size is limited to a few kilometers.
- LANs are designed to allow resources such as printer to be shared between nodes.
- It uses multipoint connections.
- Wireless LANs or WLans are new evolution of LAN technology.
- LAN mostly uses only one type of transmission media and topology.

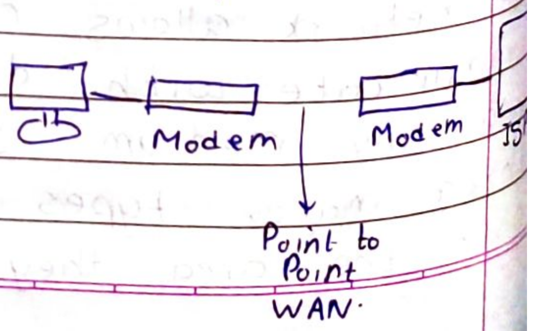


isolated Lan connecting a computer connected to a hub in a closet

2) WIDE Area Network (WAN)

- A wide area network (WAN) provides long-distance transmission of data, image, audio, and video information over large geographic areas that may comprise a country, continent or even whole world.
- ex Internet, dial-up lines
- It includes point-to-point and multipoint connections





Switched WAN

* TCP/IP Model:

- The TCP/IP model is a fundamental framework for computer network.
- It stands for Transmission Control Protocol /Internet Protocol, which are the core protocols of the internet.
- This model defines how data is transmitted over the networks, ensuring reliable communication between devices.
- It is a consive version of OSI model. which contains 5 layers. physical, data link, network, transport and application.

Application Layer.
↓ ↑
Transport Layer
↓ ↑
Network Layer
↓ ↑
Data Link Layer
↓ ↑
Physical layer.

1) Application Layer: The application layer in TCPIP is equivalent to the combined session, presentation, and application layers in the OSI model.

- It is responsible for providing network service to end users.
- Many protocols are defined in this layer like HTTP HTTPS, FTP, SMTP, POP3 etc.
- It is the top most layer and defnes how network applications should communicate.
- It handle data encoding, compression, encryption, and session management

2. Transport Layer : It ensure reliable or fast transmission of data between devices, depending on the protocol used.

- It has to ensure that data is delivered to the correct application on a device.
- Traditionally it is represented by two protocols
  - TCP (Transmission Control Protocol)
  - UDP (User Datagram Protocol).
- It is used for end-to-end communication.
- A new transport layer protocol SCTP, has been devised to meet the needs of some newer applications. (Stream Control Transmission Protocol)

3. Network Layer : The network layer (internetwork layer) supports internetworking protocol.

- It defines the protocols which are responsible for the logical transmission of data over the network.
- Main protocols in this layer are - IP, ARP, RARP, ICMP and IGMP.

- IP - Routes packets based on IP addresses (IPv4 or IPv6).
- ARP (Address Resolution Protocol) : Resolves IP addresses to MAC addresses.
- RARP (Reverse ARP) : Resolves MAC address to IP address
- ICMP (Internet Control Message Protocol) : Used for error messages and network diagnostics. (ex. ping).
- IGMP (Internet Group Management Protocol) : Used for managing multicast groups. (ex. Video streaming)

4. Data Link Layer :

- It is responsible for node-to-node communication over a physical link.

- It handles framing, addressing and error detection.
- It divides packets from Network Layer into frames for transmission.
- It uses MAC address to identify sender and receiver for each frame.
- Uses IEEE 802.2 framing and Point-to-Point Protocol (PPP) framing.

5) Physical Layer: It is the lowest layer and is responsible ~~to~~ for transmission of raw data bits over the communication medium.
- It deals with conversion of information into signals. ex digital signal into signals (electrical, radio et).
- It does not define any specific protocol.
- It defines physical medium like cables, frequencies, transmission modes, connectors etc.
- It determines transmission rate (bandwidth) topologies and modes (simplex, duplex) etc.

* Data Unit in each layer.

Message or Data (Human readable information)
↓↑
Segment or User Datagram. (Packets divided into Segments)
↓↑
Packet (Frames Segmented into packets)
↓↑
Frame (bits grouped and given MAC addresses)
↓↑
Bits (Signals like electrical, light pulse, radio).

MAC → Media Access Control.

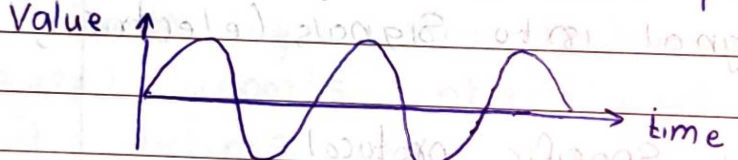
A Signal is a reprentation of data that varies over time

* Data and Signals *

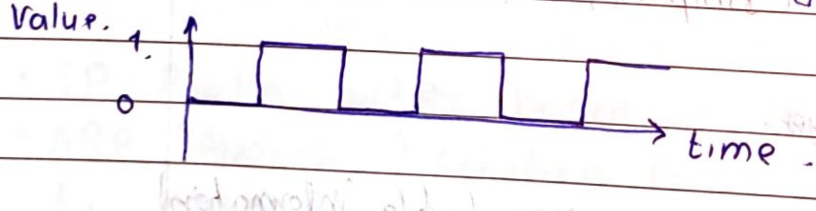
* Analog Data: It refers to information that is continuous.
It can take infinite values within a given range.

* Digital Data : It refers to information that has discrete states.
· It is represented by a sequence of binary value.

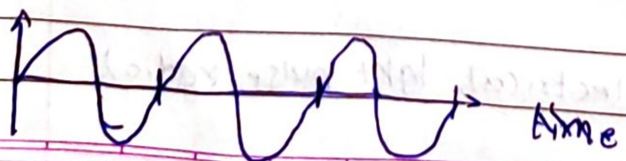
* Analog Signal: A Signal which has infinitely many levels of intensity over a period of time.
· It is the Signal which represents analog data.



* Digital Signal: A signal which represents digital data is called digital Signal.
· It has limited number of defined values (states).



* Periodic Signal: A periodic Signal completes a pattern within a measurable time frame, called a period, and repeats the pattern over Subsequent identical periods.
· The completion of one full pattern is called a cycle.



ex - Sin wave. $y(t) = A \sin(2\pi f t)$.

* Non Periodic Signal: A nonperiodic Signal changes without exhibiting a pattern or cycle that repeats over time

Value.

time

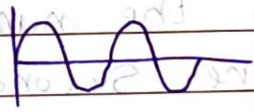
* We commonly use periodic analog Signals (because they need less bandwidth) and non periodic digital signals (because they can represent varriation in data).

* Simple Signal

A simple Signal consists of a Single frequency component and can be represented by a Sine wave or cosine wave.

ex: $S(t) = A\sin(2\pi f t + \phi)$

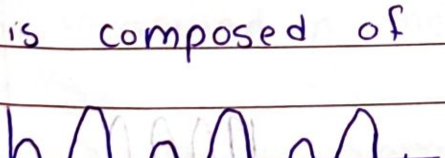
- A Simple periodic analog Signal cannot be decomposed into simpler Signals.



* Composite Signals,

It consists of multiple frequencies combined together.

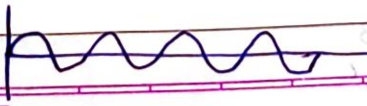
It is composed of multiple Sine waves



- first Signal is called fundamental freq, Rest are called harmonics

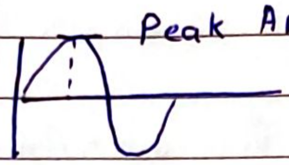
* Sine Wave:

The Sine wave is the most fundamental form of a periodic analog Signal

Value.

time

A sine wave can be represented by three parameters. the peak amplitude, Frequency and the phase.

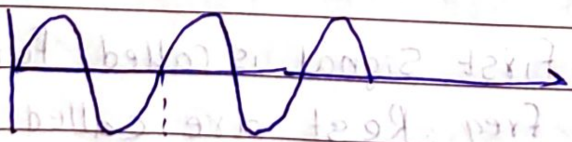
i) Peak Amplitude: It is the absolute value of its highest intensity, proportional to the energy it carries.
For electric Signals, it is measured in Volts.



Amplitude is the strength, intensity or power of a Signal. (It is the measure of how far it moved from central res position)

ii) Period and Frequency :
- Period refers to the amount of time in seconds, a Signal needs to complete 1 cycle.
- Frequency refers to the number of periods in 1s.
- It is the number of cycles a signal completes in one second.
- It is measured in Hz.

$f = \frac{1}{T}$ $T = \frac{1}{f}$



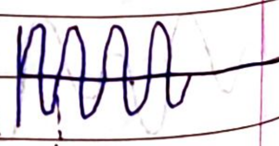
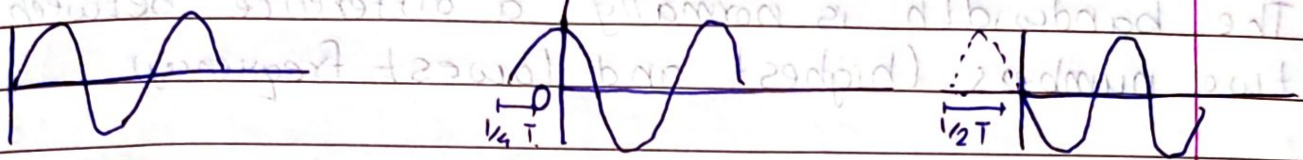


fig. Signal with frequency 12 Hz.

milli	10^{-3}	Kilo	10^{3}
micro	10^{-6}	Mega	10^{6}
Nano	10^{-9}	Giga	10^{9}
Pico	10^{-12}	Tera	10^{12}

iii) Phase :

The term phase describes the position of the waveform relative to time 0.



0 degree 90 degree 180 degree

A sine wave is offset $1/6$ cycles with rt 0. What is its phase

$\frac{1}{6} \times 360 = \underline{\underline{60^\circ}}$ $\frac{60 \times 2\pi}{360} = \frac{\pi}{3}$ rad.

iv) Wavelength :

- Wavelength depends on both frequency and medium.
- Wavelength is the distance a simple signal can travel in one period

(A) Wavelength = propagation speed × period = $\frac{\text{Speed}}{\text{frequency}} = \frac{V}{f}$

- It is measured in meters (usually).

If a composite signal is periodic, the decomposition gives a series of signals with discrete frequencies, if the composite signal is non periodic, the decomposition gives a composition of sine waves with continuous frequencies.

Composite Periodic Signal.

v) Bandwidth

- The range of frequencies contained in a composite Signal is its bandwidth.
- The bandwidth is normally a difference between two numbers. (highest and lowest frequency).

$$\text{Bandwidth}_{(analog)} = f_{max} - f_{min}.$$

ex. A radio Station transmitting between 88 and 108 MHz has a bandwidth of 20 MHz.

- For digital Signals, bandwidth refers to the rate at which data is transmitted.
- It is the amount of data transferred over a communication channel per unit of time, measured in bits per Second.

$$\text{Bandwidth (digital)} = \text{Data Rate} = \frac{\text{Total data}}{\text{Time}}.$$

ex. A wifi network with bandwidth 100 Mbps can transfer 100 million bits per second.

vi) Instantaneous Amplitude:

Instantaneous amplitude refers to the amplitude (strength or value) of a wave at any specific point in time. Instantaneous amplitude vary at each point in time depending on the wave's position.

$$A(t) = A \sin(2\pi f t + \phi)$$

* Signal-to Noise Ratio (SNR)

$$SNR = \frac{\text{average Signal Power}}{\text{average noise power.}}$$

$$\boxed{\begin{array}{l} 1)\ SNR_{dB} = 10 \log_{10} SNR \\ 2)\ SNR = 10^{\frac{SNR_{dB}}{10}} \end{array}}$$

* Noiseless Channel: Nyquist Bit Rate.

For a noiseless channel, the Nyquist bit rate formula defines the theoretical maximum bit rate.

$$BitRate = 2 \times bandwidth \times \log_2 L$$

L = Levels (No of Signal levels)

- Increasing the levels of a Signal reduces the reliability of the System
- ex, if $L=2$ we can easily distinguish between 0 and 1.
- But if $L=64$, the receiver must be very Sophisticated to distinguish between 64 different levels.
- $2^{bitrate}$ = No of Voltage Levels.
- No of bits = $\log_2$ (No of Voltage Levels)

$\log_2(64) = 6 \Rightarrow$ We can transfer 6 bit at a time if we have 64 Voltage levels.

* Noisy Channel: Shannon Capacity:

$$\text{Capacity} = \text{bandwidth} \times \log_2(1 + SNR)$$

* Transmission Impairment:

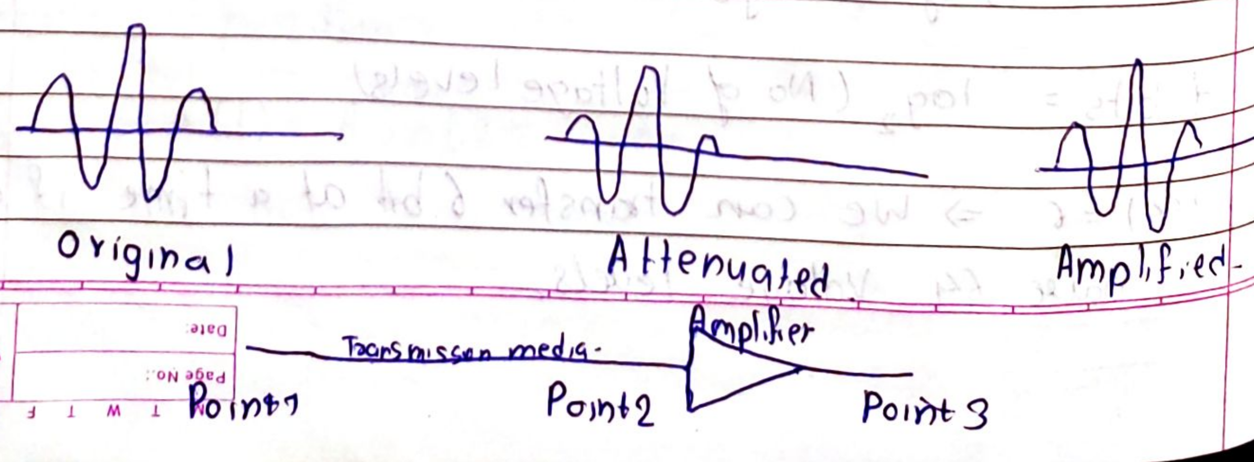
- Signals travel through transmission media, which are not perfect.
- The imperfection causes signal impairment.
- This means signal at the beginning of the medium is not the same as the signal at the end of the medium.
- What is sent is not what is received.

Transmission Impairment

attenuation. distortion. noise.

i) Attenuation: Attenuation means a loss of energy. When a signal, simple or composite, travels through a medium, it loses some of its energy in overcoming the resistance of the medium.

- That is why a wire carrying electrical signals get warm.
- After this loss amplifiers are used to amplify the signal.



Decibel: To show signal has lost or gained strength, we use the unit decibel.

- The decibel (dB) measures the relative strengths of two signals or one signal at two different points.

$$dB = 10 \log_{10} \frac{P_2}{P_1} = 20 \log_{10} \left(\frac{V_2}{V_1}\right)$$

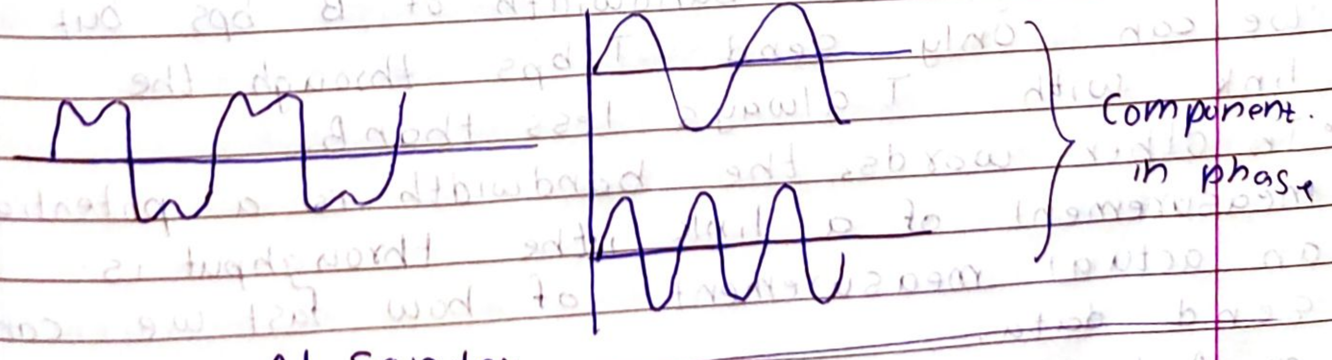
decibel is negative if a signal is attenuated and positive if a signal is amplified.

P_1, P_2 are power of signal at point 1, 2 respectively.

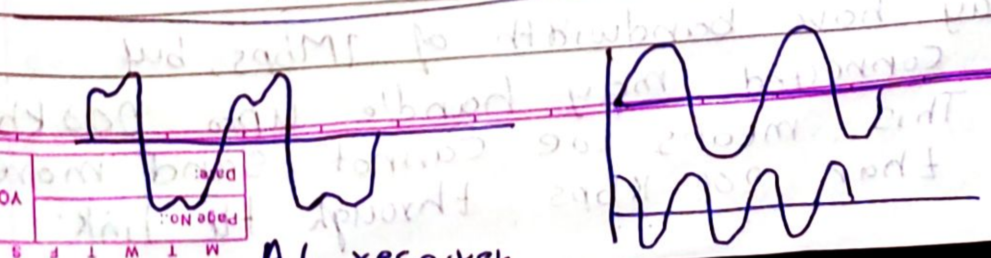
ii) Distortion: Distortion means that the signal changes its form or shape.

- Distortion can occur in a composite signal made of different frequencies.
- Each signal component has its own propagation speed through a medium and, therefore, its own delay in arriving at the final destination.

Difference in delay may cause a difference in phase if the delay is not exactly the same.



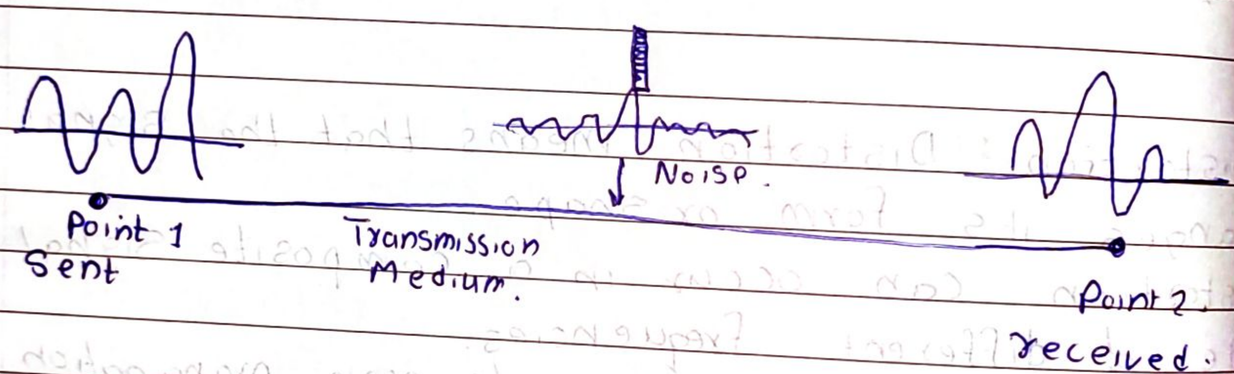
At sender



At receiver

iii) Noise:

- Several types of noise, such as thermal noise, induced noise, cross talk and impulse noise, may corrupt the signal.
- Thermal noise is the random motion of electrons in a wire which creates an extra signal not originally sent by the transmitter.
- Induced noise comes from sources such as motors and appliances.
- Crosstalk is effect of one wire on other.
- Impulse noise is a spike that comes from power lines, lightening etc.



★ Throughput:

- The throughput is a measure of how fast we can actually send data through a network.
- It seems similar to bandwidth in bit per second but is different.
- A link may have bandwidth of B bps, but we can only send T bps through the link with T always less than B.
- In other words, the bandwidth is a potential measurement of a link, the throughput is an actual measurement of how fast we can send data.

ex. A link may have bandwidth of 1Mbps, but the device connected may handle upto 200 kbps. This means we cannot send more than 200 kbps through the link.